Worksheet 3: Agent Architect

Design, build, and ship a real AI agent

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Building an agent isn't just coding. It's deciding WHAT problem to solve, WHO it's for, WHAT tools it needs, and WHERE a human should stay in the loop. This worksheet takes you from idea to deployed product -- and forces you to think about the decisions most people skip.

A. The problem (not the technology)

Start with a real problem. Don't think 'I want to build an agent.' Think 'I have a problem that an agent could solve.'

Who has this problem? (be specific -- not 'students' but '2nd year CSE students preparing for placements'):

What task is painful or slow for them?

How do they currently handle this task? (the workaround)

What goes wrong with the current approach?

B. Does this actually need an agent?

Not every problem needs AI. Be honest.

Solution type	Would this work?

A better Google Doc or spreadsheet	
A simple script (no AI)	
A chatbot with a good prompt (Level 2)	
RAG -- AI over specific documents (Level 3)	
Agent with tools (Level 4)	

My honest assessment -- the simplest solution that would actually work:

If a spreadsheet solves it, use a spreadsheet. Using AI where it isn't needed is just as bad as not using it where it is.

C. Agent design

If an agent (or RAG + agent) is the right approach, design it here.

The Brain (LLM)

System prompt -- how should the agent behave? (write the full prompt):

The Tools

List every tool your agent needs. For each, describe what it does and what input it takes.

Tool name + description	Input / Output
Tool 1:	
Tool 2:	
Tool 3:	

The Memory (if using RAG)

What documents or data go into the knowledge base?

What should RAG handle vs. what should tools handle?

D. The human in the loop

Every responsible AI system has boundaries. Where does the human stay in control?

What decisions should the agent make on its own?

What decisions must a human review or approve?

What happens if the agent is WRONG? What's the worst case?

How would a user know the agent made a mistake?

E. Build log

Track what you built, what worked, and what didn't.

What I tried	What happened

Hardest technical problem I hit and how I solved (or didn't solve) it:

F. Ship it

An agent that isn't deployed is a homework assignment. An agent with a URL is a product.

My deployment method (Dify URL / Gradio public link / other):

My public URL:

I tested it with someone else. Their reaction:

G. The 2-minute pitch

Structure your hackathon demo.

The problem (1 sentence):

Who it's for (1 sentence):

How it works -- the 3-step explanation (what tools, what data, what flow):

Live demo -- what question will you ask to show it off?

H. Reflection: what kind of builder are you?

This workshop gave you tools. What matters is what you do with them after today.

The moment during the workshop when something 'clicked' for me:

The thing I built today that I'm most proud of:

One real problem in my life I could solve with what I learned today:

My next step -- what I will build or learn in the next 7 days:

You walked in as a consumer. You're walking out as a builder.
The AI won't take your job. But someone who builds with AI might.
Be that someone.